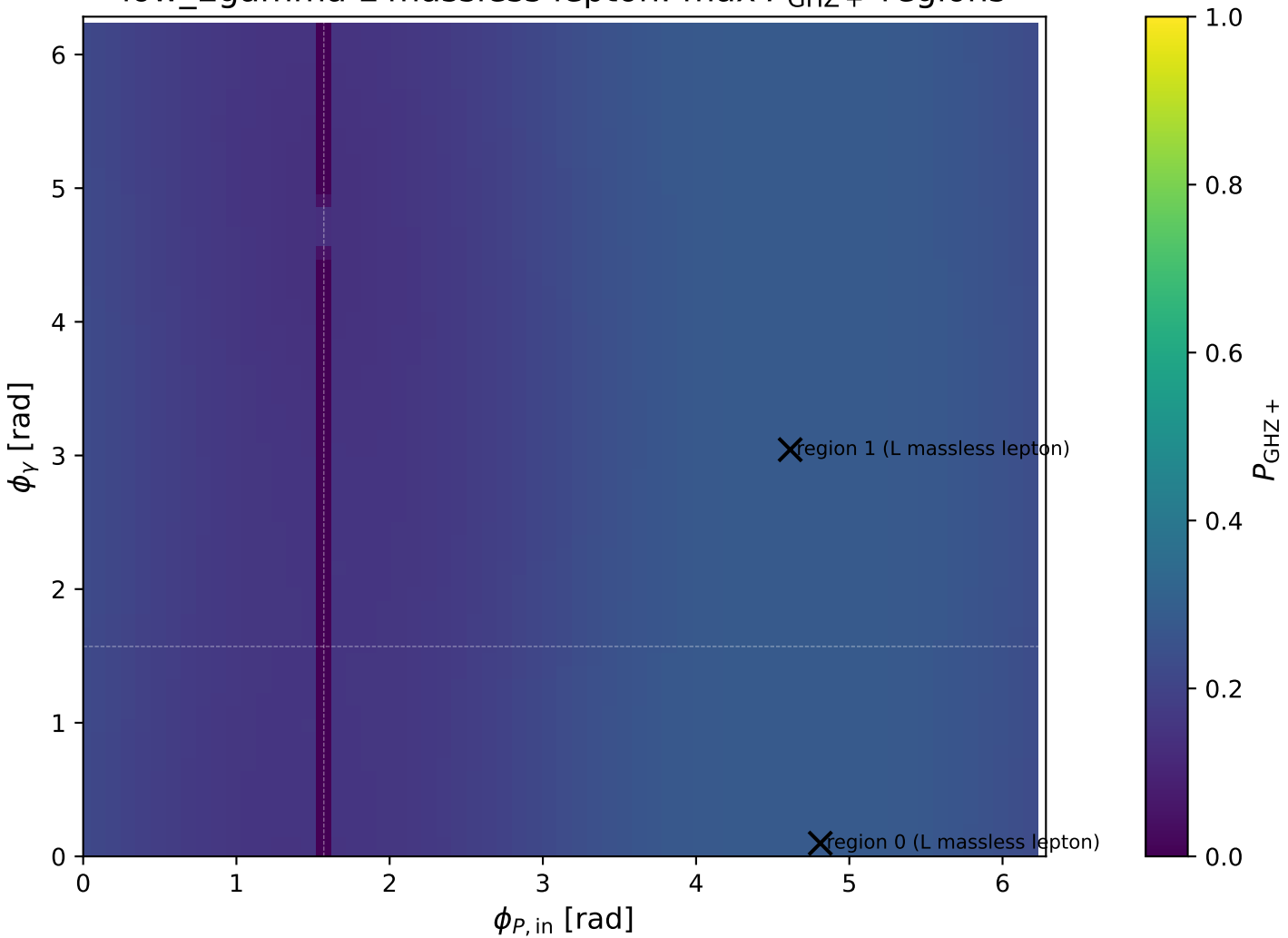
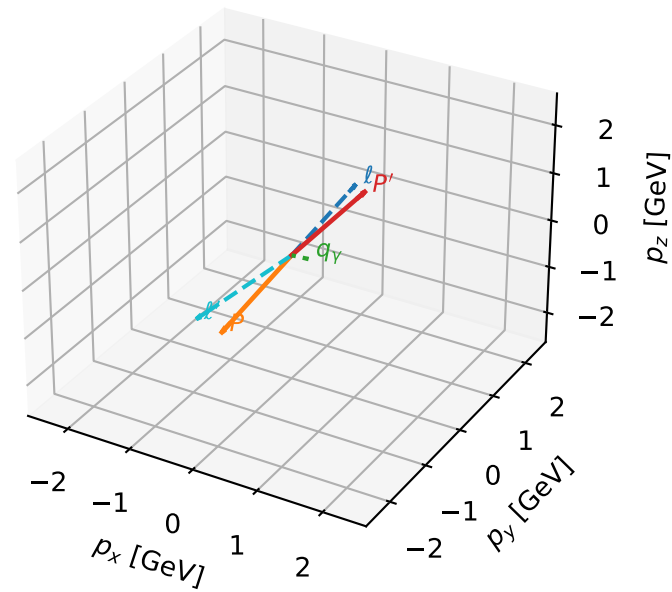


low_Egamma L massless lepton: max $P_{\text{GHZ}+}$ regions



L massless lepton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

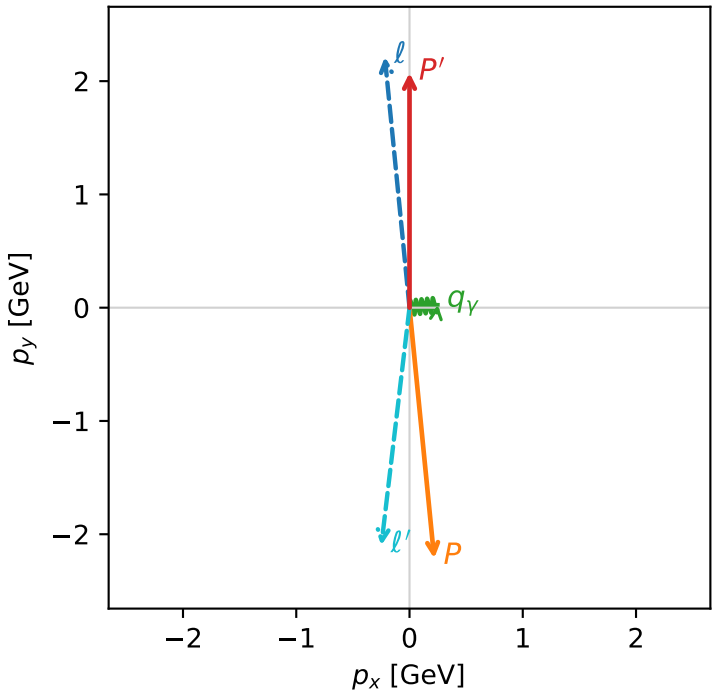


ghz_purity_low_egamma_l_lepton_region_0 $P_{\text{GHZ+}}=0.27971$
region: low_Egamma_L_lepton_region_0
outgoing-state purity: 0.999864
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{l_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.07351$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.66897$, $\phi_P=4.81056$
 $E_V=0.25$, $\phi_V=0.0981748$
 $Q^2=18.5794$, $x_B=0.947171$, $t=-18.4086$, $W^2=1.91613$, $y=0.950556$
 $\theta(l', \gamma)=102.388$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -0.2180$ $p_y= 2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.2180$ $p_y= -2.2137$ $p_z= 0.0000$
 l' $E= 2.1127$ $p_x= -0.2488$ $p_y= -2.0980$ $p_z= 0.0000$
 P' $E= 2.2758$ $p_x= 0.0000$ $p_y= 2.0735$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2488$ $p_y= 0.0245$ $p_z= 0.0000$

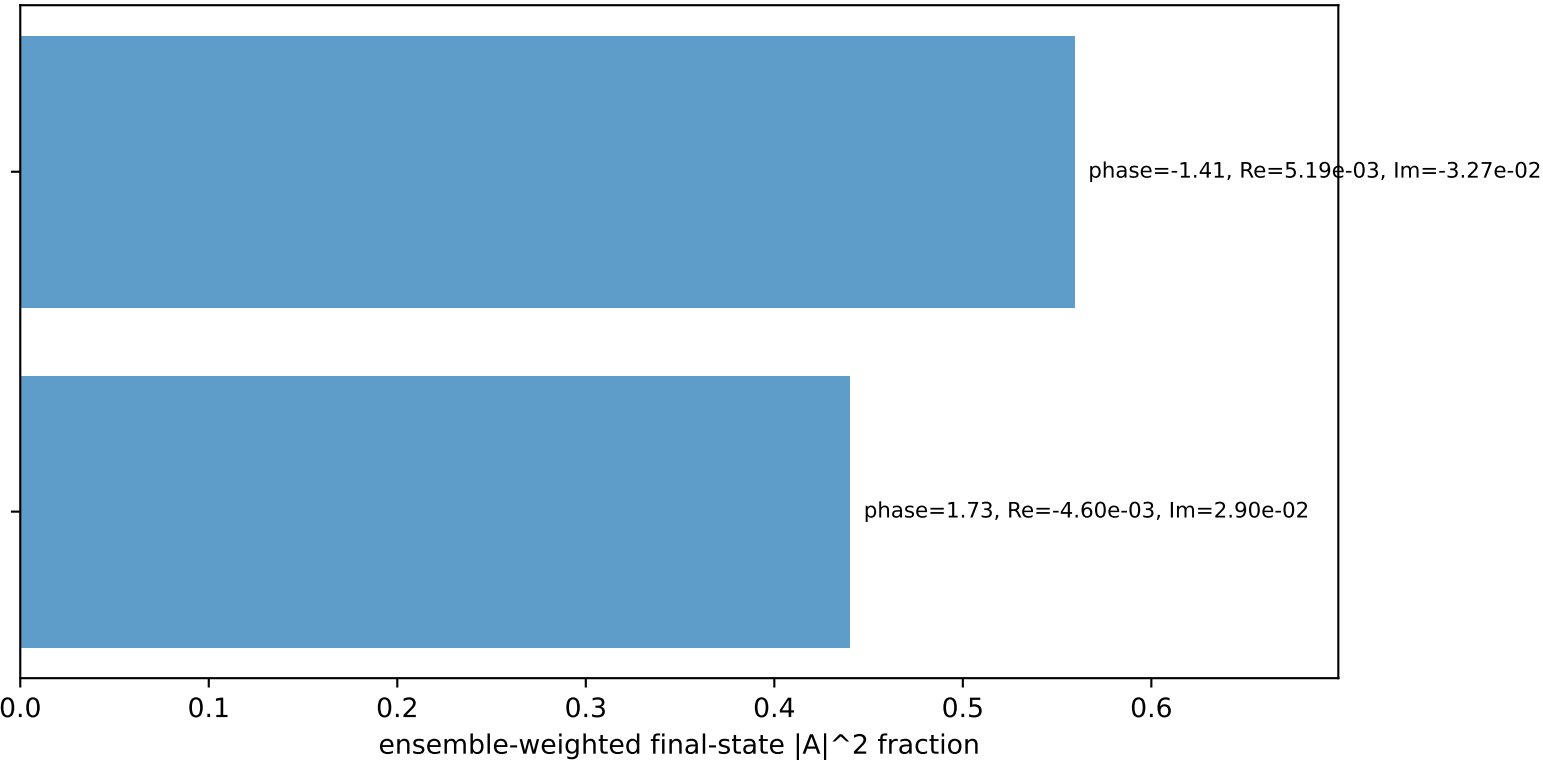
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron L, proton h=+1

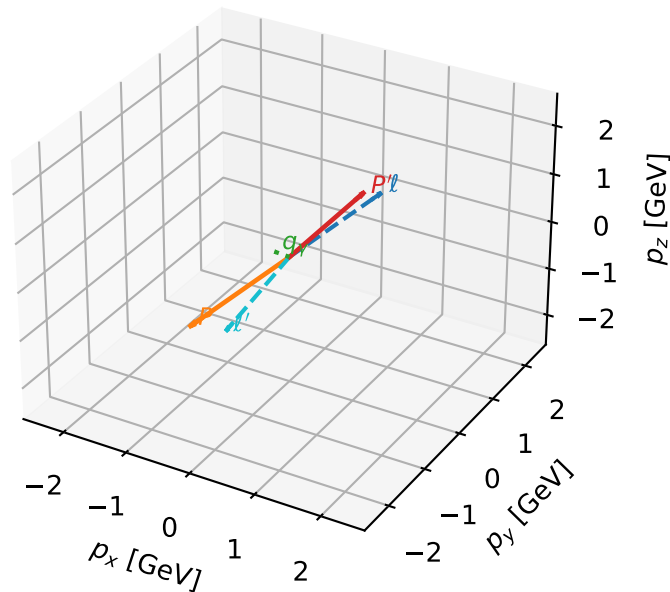
$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton h=+1

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L massless lepton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

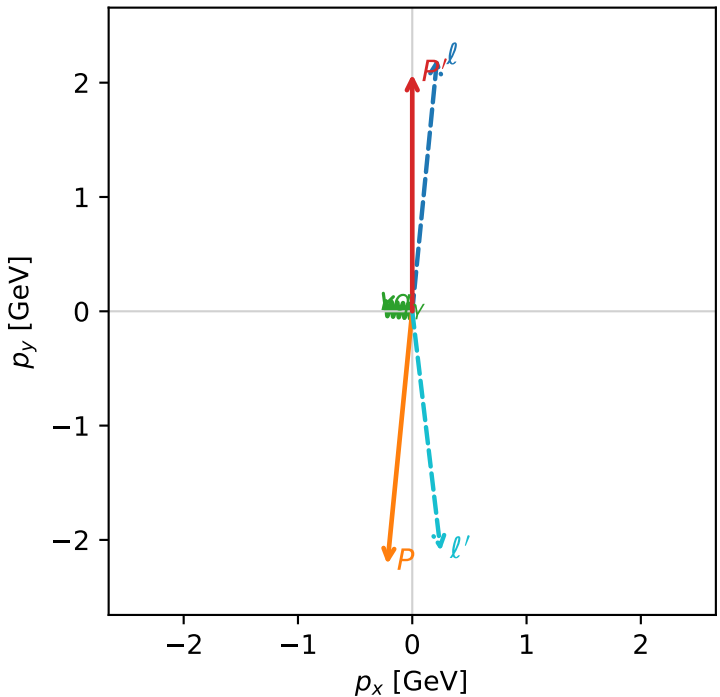


ghz_purity_low_egamma_l_lepton_region_1 $P_{\text{GHZ+}}=0.27971$
region: low_Egamma_L_lepton_region_1
outgoing-state purity: 0.999864
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{t_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.07351$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.47262$, $\phi_p=4.61421$
 $E_\gamma=0.25$, $\phi_\gamma=3.04342$
 $Q^2=18.5794$, $x_B=0.947171$, $t=-18.4086$, $W^2=1.91613$, $y=0.950556$
 $\theta(l', \gamma)=102.388$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 0.2180$ $p_y= 2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.2180$ $p_y= -2.2137$ $p_z= 0.0000$
 l' $E= 2.1127$ $p_x= 0.2488$ $p_y= -2.0980$ $p_z= 0.0000$
 P' $E= 2.2758$ $p_x= 0.0000$ $p_y= 2.0735$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2488$ $p_y= 0.0245$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron L, proton $h=+1$

$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton $h=+1$

Leading initial-ensemble/final-state components (N=2, retained=100.0%)

